

Foundations of Cybersecurity

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About

- I am:
 - a Research Scientist at the **National Research Council of Italy**
 - started as a networking guy, now in cybersecurity
 - held courses for Ph.D. and M.Sc. students on security and computer networks
 - my main research areas are **information hiding** and **software security**.
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Goals

- The **course** has the following **goals**:
 - provide a comprehensive introduction to **basic cybersecurity** concepts
 - outline the most common **attack surfaces**
 - give a background for **autonomously** gathering information
 - show some **toy examples**
 - **have fun!**
- This course is **not** about:
 - hacking (read *Phrack Magazine*, <https://phrack.org>)
 - hardware security and cryptography
 - things likely to fall outside your early career as an engineer.

Grading

- There are **two paths** to the **final exam**.
- The **First Path** (**strongly suggested**) is organized in two steps.
- **Oral** presentation:
 - a short seminar of 15-20 minutes for showing results of an assignment
 - groups of students (size depends on the number of attendees)
 - prepare supporting material and slides you believe are useful
 - results will become part of the course
 - score: up to **24/30**.
- **Written** test:
 - **three** open questions on (very) core concepts
 - **mandatory** even if you have already scored 18/30 in the oral presentation
 - score: **3 per question**.
- **Final grade**:
 - if total score > 31, then “30 with honors”
 - else the plain sum.

Grading

- The **Second Path** is organized in a single step.
- **Written test:**
 - **eleven** open questions on (very) core concepts
 - score: **3 per question**.
- **Final grade:**
 - **if** total score > 31, **then** “30 with honors”
 - **else** the plain sum.

Other Information

- Checkpoint:
 - I propose a **mid-term checkpoint**
 - simulation of a written test
 - the outcome **does not contribute in any manner** to the final evaluation.
- I am a very open-minded, but **please do not**:
 - copy
 - steal work
 - copy/paste concepts and data without proper attribution
- On the use of AI/LLMs:
 - ChatGPT & Co. are welcome if used to **support** routinary tasks
 - not allowed if they **prevent** you from **learning**!

Course Outline

- The course is organized in:
 - **6** modules covering major aspects of cybersecurity
 - **1** final wrap-up module
 - **1** seminar held by an expert (**tentative proposal**).

1. Introduction and Basics:

- motivations on the importance of cybersecurity
- the Cyber Kill Chain
- attack surfaces and attack surface reduction practices
- hints at the human element.

2. Security Analysis and Modeling:

- common weaknesses enumeration
- common vulnerability enumeration
- common vulnerability scoring system
- testing strategies.

Course Outline

3. Software Supply Chain Security:

- main attack entry points
- dependency attacks
- (typo | slop | combo) squatting
- starjacking and dependency confusion
- build system attacks and reproducible builds.

4. Malware:

- major threats and architectures
- malware analysis, packers and anti-forensics
- YARA rules
- binary and source code obfuscation and minification.

5. Network Security:

- major network attack types
- sniffing, spoofing and DoS/DDoS
- firewalls, policy enforcing and hardening
- network address translation, honeypots, and segmentation.

Course Outline

6. Information Hiding:

- information hiding and steganography
- network covert channels
- local covert channels and colluding applications
- sanitization (including side-channel attacks)
- watermarking of data, software and AI models.

7. Conclusions:

- discussion of presentations (**tentative placement**)
- final recommendations and lessons learned.

A Remark on AI and LLMs

- This is **not** a course on AI security.
- However, it is undeniable that AI/LLMs:
 - allow to **produce new software** at an unprecedented pace
 - automatize **discovering vulnerabilities** and **generating patches**
 - **summarize logs** and “explain” suspicious activities
 - **analyze** binaries and **de-obfuscate** code
 - change the **speed** at which **attacks** are developed
 - introduce **new attack surfaces** (e.g., prompts injection and refusals)
 - **lower the barrier** for preparing **attacks** or **phishing** campaigns
 - ...
- Then, when relevant I will briefly discuss:
 - the impact of AI on standard cybersecurity routines
 - new threats and risks.

Course Material

- There is **no** a **unique** book that can be read cover to cover.
- Suggested sources:
 - slides
 - examples and open repositories
 - official or selected websites
 - introductive research papers.
- The needed **material** will be published in the GitHub of the course:
 - <https://github.com/lucacav/foc>
 - clone the repository and stay synced (*gh repo clone lucacav/foc*).
- Disclaimer:
 - this is the **second edition** of “Foundations of Cybersecurity”
 - material for the past edition has been packed into a release (i.e., 2025/2026)
 - **comments and suggestions are welcome.**